

Quasi-Borel Spaces (Cont. Closed categories)

Notation:

• $f \circ g = g \circ f$

• $\mathcal{C}$ category, $\mathcal{C}(X, Y)$ morphism set $X \rightarrow Y$.

1) Cartesian Closed Categories (CCC).

vs Curry - Howard - Lambek

~~Category~~ CCC

Logic

PL

obj. $\longleftrightarrow$
mem. $\longleftrightarrow$

prop.
proof

$\longleftrightarrow$ types
 $\longleftrightarrow$ functions

~~Def~~ Def CCC: $\mathcal{C}$ category. $\mathcal{C}$ is CCC:

1) $1 \in \mathcal{C}$

2) $\forall X, Y \in \mathcal{C}$ have $X \times Y \in \mathcal{C}$

3) $\forall X, Y \in \mathcal{C}$ have $Y^X \in \mathcal{C}$:

Satisfy:

$$\mathcal{C}(X \times Y, Z) \cong \mathcal{C}(X, Z^Y)$$

Second: Means we have higher Order functions

Note: Think of $1 \rightarrow X$ as "picking element of X "

$$\# \mathcal{P}(X, Y) \cong \mathcal{P}(1 \times X, Y) \cong \mathcal{P}(1, Y^X)$$

Ex 1: Set is CCC.

$$1 = \mathbb{S} * \mathbb{S}$$

$$Y^X = \text{Set}(X, Y)$$

$$X \times Y \rightarrow Z \quad \text{same as} \quad X \rightarrow (Y \rightarrow Z)$$

Ex 2: Consider a category $\mathcal{L}$ of propositions and (exist of) proofs.

Ob: Propositions $\alpha, \beta, \dots$

$$\text{Morphisms: } \mathcal{L}(\alpha, \beta) = \begin{cases} \mathbb{S}^{\times \mathbb{N}} & \text{if } \alpha \vdash \beta \\ \emptyset & \text{otherwise} \end{cases}$$

$\mathcal{L}$ is CCC if:

$$* \top \in \mathcal{L}$$

$$* \forall \alpha, \beta \text{ have conj } \alpha \wedge \beta$$

$$* \forall \alpha, \beta \text{ have impl } \alpha \Rightarrow \beta$$

Curryying:

$$\alpha \wedge \beta \vdash \gamma \quad \text{same as} \quad \alpha \vdash \beta \Rightarrow \gamma$$

Problem: Meas is not CCC

In particular: $\text{Meas}(\mathbb{R}, \mathbb{R})$, and

$$\text{ev}: \text{Meas}(\mathbb{R}, \mathbb{R}) \times \mathbb{R} \rightarrow \mathbb{R}$$
$$f, r \mapsto f(r)$$

Is never a meas. func.

$\rightarrow$ Even $P(\text{Meas}(\mathbb{R}, \mathbb{R})) \otimes \Sigma_{\mathbb{R}}$ does not work.

2) QBS

For now fix some "sample space" $(\Omega, \Sigma_{\Omega})$.
 $(\mathbb{R}, \Sigma_{\mathbb{R}})$

Idea: A random variable is a meas. set

$$(\Omega, \Sigma_{\Omega}) \xrightarrow{\text{meas. } f} (X, \Sigma_X)$$

Def: QBS (category of Quasi-Borel spaces) is category with objects pairs $(X, \mathcal{M}_X)$ where X is a set and $\mathcal{M}_X \subseteq \text{Set}(\Omega, X)$, subject to condition:

Q1

(1) If $\alpha \in \mathcal{M}_X$ and $f: \Omega \rightarrow \Omega$ is a meas. func, then $f_* \alpha \in \mathcal{M}_X$.

(2) All constant $\alpha: \Omega \rightarrow X \in \mathcal{M}_X$

(3) If $\Omega = \biguplus_{i \in \mathbb{N}} S_i$, each S_i measurable, and we have

$\alpha_1, \alpha_2, \dots \in \mathcal{M}_X$, then $\beta \in \mathcal{M}_X$, where

$$\beta(r) = \alpha_i(r) \text{ if } r \in S_i$$

Then a morphism in QBS

$$f: (X, \mathcal{M}_X) \longrightarrow (Y, \mathcal{M}_Y)$$

is a function $f: X \rightarrow Y$, that "respects $\mathcal{M}$." That means

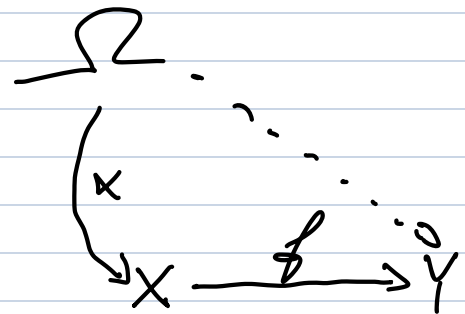
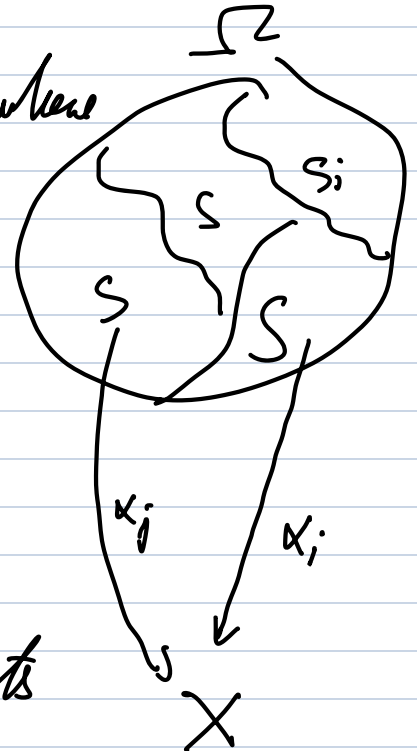
$$\forall \alpha \in \mathcal{M}_X, \quad \alpha \circ f \in \mathcal{M}_Y$$

Ex:

- $(\mathbb{R}, \mathcal{M}_{\mathbb{R}})$, where $\mathcal{M}_{\mathbb{R}} = \text{Meas}(\mathcal{B}(\mathbb{R}, \Sigma_{\mathbb{R}}), (\mathbb{R}, \Sigma_{\mathbb{R}}))$

$$(\Omega, \mathcal{M}_{\Omega}) \longleftrightarrow \Sigma_{\Omega}$$

$$f \longrightarrow \Sigma_{\perp}, T\Sigma$$



Def: For $(X, \mathcal{M}_X) \in \text{QBS}$, a prob. measure on X is a pair (α, μ) where $\alpha \in \mathcal{M}_X$ and $\mu : \Omega \rightarrow [0, 1]$ is a prob measure (in usual sense)

Remark: Let (α, μ) be ~~prob~~ prob meas on $(X, \mathcal{M}_X)$

- Pushforward: Let $f : (X, \mathcal{M}_X) \rightarrow (Y, \mathcal{M}_Y)$
Then $(\alpha \circ f, \mu)$ is prob on Y .

- Integrate: Let $f : (X, \mathcal{M}_X) \rightarrow (\mathbb{R}, \mathcal{M}_{\mathbb{R}})$
And prob. meas. (α, μ) on $(X, \mathcal{M}_X)$.

Then we can integrate:

$$\int f d(\alpha, \mu) \triangleq \int_{\mathbb{R}} (\alpha \circ f) d\mu$$

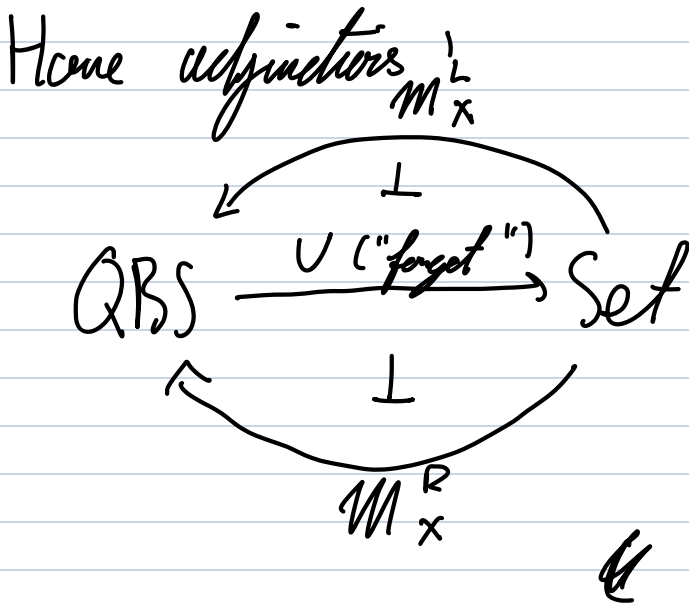
$$\alpha \in \mathcal{M}_X, \quad \alpha \circ f \in \mathcal{M}_{\mathbb{R}}$$

Adjunctions:

Consider set X . How to equip with $(X, \mathcal{M}_X)$?

Max choice: $\mathcal{M}_X^{\mathbb{R}} = \text{Set}(\Omega, X)$

Min choice: $M_x^L = \text{"Minimized thing generated by (1)-(3)"}$
 $\approx$ "Locally constant on 'certain' partitions of Ω " Ω



$$U(x, M_x) = X$$

i.e.

$$QBS((x, M_x^L), (y, M_y)) = Set(X, Y) = QBS((x, M_x), (y, M_y^R))$$

Meas: Remember: Want M_x Meas QBS
 to be "meas. function $\Omega \rightarrow X$ "

We have $(X, \Sigma_X) \in \text{Meas}$.

Define

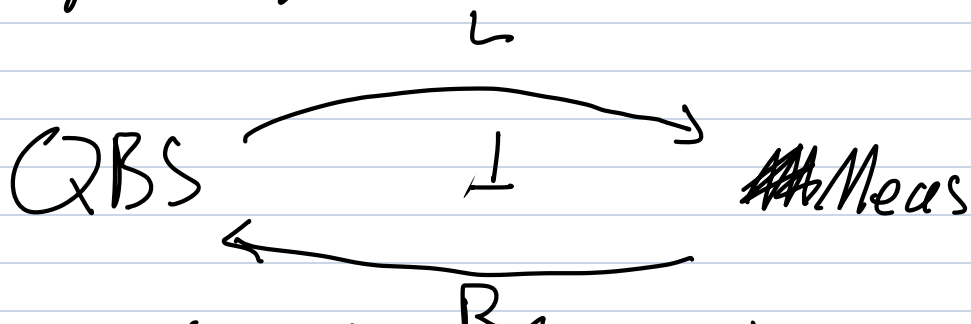
$$M_{\Sigma_X} = \text{Meas}((\Omega, \Sigma_\Omega), (X, \Sigma_X))$$

Other meas: Say we have $(Y, M_Y) \in \text{QBS}$.

Biggest we can make Σ_Y is

$$\Sigma_{M_Y} = \{U \subseteq Y \mid \forall x \in M_Y. \quad x^*(U) \in \Sigma_\Omega\}$$

Fact/Props: Adjunction



Where $L(X, M_X) = (X, \Sigma_{M_X})$ and
 $R(Y, \Sigma_Y) = (Y, M_{\Sigma_Y})$.

That is:

$$\begin{aligned}
 & \text{QBS}(X, M_X), (Y, M_{\Sigma_Y}) \\
 &= \text{Meas}(X, M_{\Sigma_X}, (Y, \Sigma_Y))
 \end{aligned}$$

$\leadsto$ R is fully faithful (only for X, Y Boolean Stone set)

$$\begin{aligned}
 & \text{Meas}(X, \Sigma_X), (Y, \Sigma_Y) \\
 &= \text{QBS}(X, M_{\Sigma_X}, (Y, M_{\Sigma_Y}))
 \end{aligned}$$

3 CCC: $(X, M_X), (Y, M_Y)$
 $\cancel{Y}^x Y^x = (\text{QBS}(X, Y), M_{Y^x})$

Where $M_{Y^X} = \{ \alpha: \Omega \rightarrow Y^X \mid \text{univ}(\alpha): \Omega \times X \rightarrow Y \}$

Primer 19

$R: \text{Meas} \Rightarrow \text{QBS}$